



School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Major Project - Abstract Form

STUDENT DETAILS:

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TITLE & ABSTRACT:

Title: AI-Driven Waste Management System for Sustainable Resource Optimization

Abstract - Waste management has become one of the most pressing challenges in the pursuit of environmental sustainability and urban resilience. Traditional methods of waste segregation are often manual, time-consuming, and prone to inaccuracies, leading to inefficient recycling and increased landfill usage. To address these limitations, this project proposes an **AI-Driven Waste Management System for Sustainable Resource Optimization**. The system leverages advanced deep learning techniques and computer vision to automatically classify waste into predefined categories such as recyclable, organic, hazardous, and non-recyclable materials. By integrating edge-based AI detection with cloud-enabled dashboards, the solution ensures real-time monitoring and efficient decision-making for waste management authorities. The proposed framework not only enhances classification accuracy but also contributes to **resource optimization** by providing data-driven insights on recycling rates, energy savings, and carbon footprint reduction. A curated dataset of waste images will be used to train convolutional neural networks (CNNs) and transfer learning models such as MobileNet and ResNet to achieve robust performance across varied environmental conditions. The system will also incorporate sustainability analytics to track key metrics including waste diversion from landfills, CO₂ emissions avoided, and energy conserved through recycling. Furthermore, a user-friendly dashboard and chatbot interface will provide interactive visualization and awareness tools for stakeholders, ranging from municipal bodies to the general public. Ultimately, this project aims to support the goals of the **circular economy** by maximizing resource recovery, minimizing environmental impact, and enabling smarter decision-making in waste management practices.

Keywords - AI-driven waste management, deep learning, computer vision, sustainable resource optimization, waste classification, recycling efficiency, circular economy, transfer learning

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